

PROJECT DOCUMENTATION:

Project Title:

Automated CI/CD Pipeline using Jenkins, Docker, and AWS EC2

Objective:

To build a fully automated CI/CD pipeline that:

- Pulls code from GitHub
 - Builds a Docker image
 - Deploys the application on Amazon EC2
 - Automatically triggers on every code push
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Architecture:

GitHub → Jenkins → Docker → EC2 → Browser

Tools & Technologies Used:

- Jenkins → CI/CD automation
 - Docker → Containerization
 - Amazon EC2 → Cloud hosting
 - GitHub → Source code repository
 - Python → Application
-

Workflow:

Code Management

- Application code stored in GitHub
- Includes:
 - app.py

- Dockerfile
 - requirements.txt
-

2 Jenkins Pipeline

- Jenkins job created (pipeline-project)
 - Pipeline script includes:
 - Clone repository
 - Build Docker image
 - Run container
-

3 Build Process

`docker build -t pipeline-app .`

4 Deployment

`docker run -d -p 5000:5000 pipeline-app`

5 Automation (Webhook)

- GitHub webhook configured
 - Jenkins automatically triggers on push
-

6 Application Access

`http://<Elastic-IP>:5000`

Key Features:

- ✓ Fully automated CI/CD pipeline
- ✓ Webhook-based trigger (no manual build)

- ✓ Containerized deployment
 - ✓ Live application hosted on AWS
 - ✓ End-to-end DevOps workflow
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Challenges Faced:

1 Jenkins crash due to disk space

- Error: /tmp space below threshold
- Jenkins node went offline

Solution:

- Cleared unnecessary files
 - Adjusted disk thresholds
-

2 Jenkins HTTP 500 error

- Cause: Cache corruption (/var/cache/jenkins)

Solution:

```
sudo rm -rf /var/cache/jenkins  
sudo systemctl restart jenkins
```

3 Git not installed in EC2

- Pipeline failed during clone

Solution:

```
sudo dnf install git -y
```

4 Branch mismatch issue

- Error: Couldn't find any revision to build

Solution:

- Updated branch to main
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5 Docker permission issue

- Jenkins couldn't access Docker

Solution:

```
sudo usermod -aG docker jenkins  
sudo chmod 666 /var/run/docker.sock
```

6 Dynamic Public IP issue

- EC2 IP changed after restart
- Webhook stopped working

Solution:

- Assigned Elastic IP
-

7 Webhook not triggering initially

- No changes pushed → no trigger

Solution:

- Created new file
 - Verified webhook delivery (200 OK)
-

8 Windows command confusion

- Used touch command (Linux only)

Solution:

- Used CMD command:

```
echo test > test.txt
```

□ Learning Outcomes

- CI/CD pipeline design
- Jenkins automation
- Docker containerization
- AWS EC2 deployment
- Webhook integration
- Debugging real-world issues

Pipeline script:

```
pipeline {  
    agent any  
  
    stages {  
  
        stage('Clone Code') {  
            steps {  
                git branch: 'main', url: 'https://github.com/MohamedRiyaz12/pipeline-project.git'  
            }  
        }  
  
        stage('Build Docker Image') {  
            steps {  
                sh 'docker build -t pipeline-app .'            }  
        }  
  
        stage('Run Container') {
```

```
steps {  
  sh ""  
  
  docker stop pipeline-app || true  
  
  docker rm pipeline-app || true  
  
  docker run -d -p 5000:5000 --name pipeline-app pipeline-app  
  ""  
}  
}
```

}

}

git branch: 'main', url: '...' – Clone code

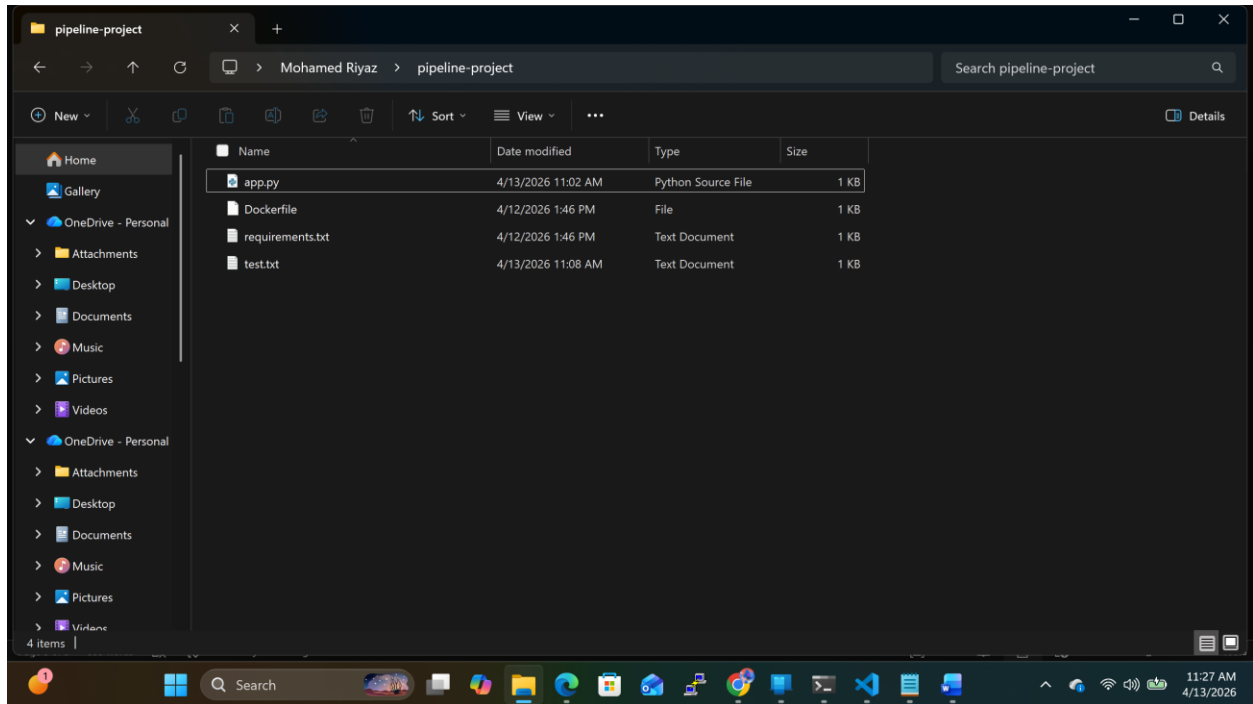
docker build -t pipeline-app . – Build Docker image

docker stop pipeline-app || true – Stops old container

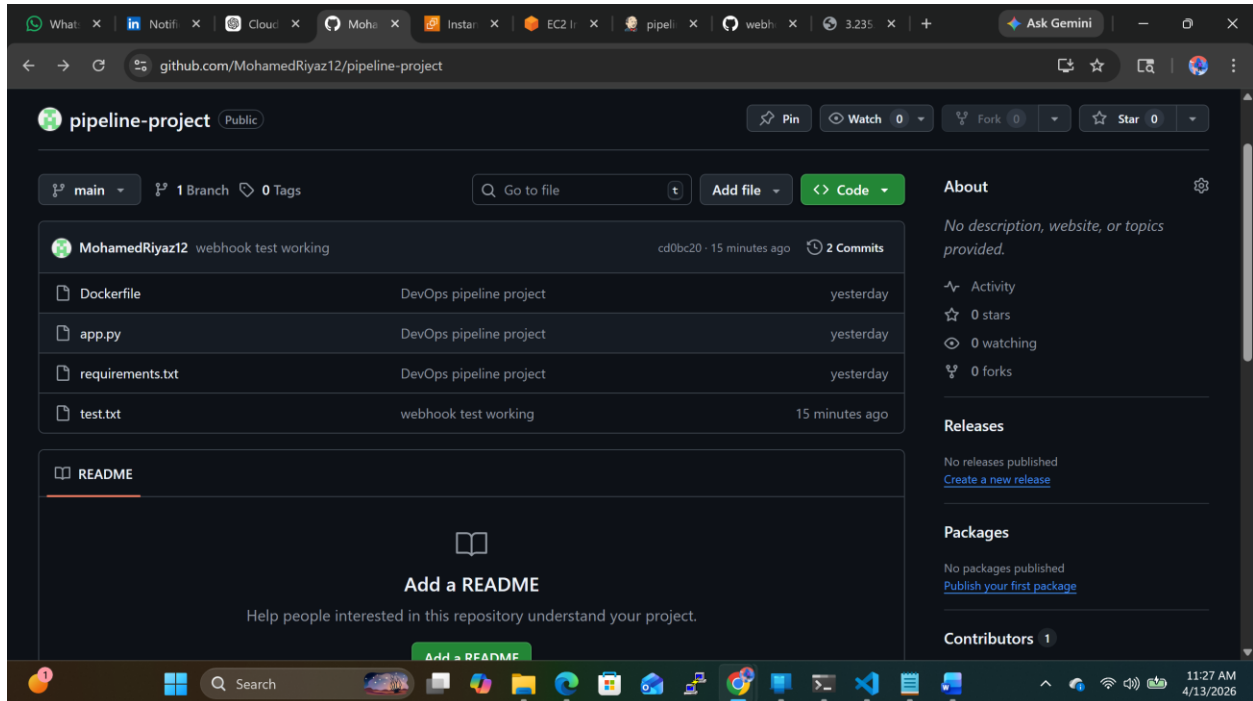
docker rm pipeline-app || true – Removes old container

docker run -d -p 5000:5000 --name pipeline-app pipeline-app – Runs new container

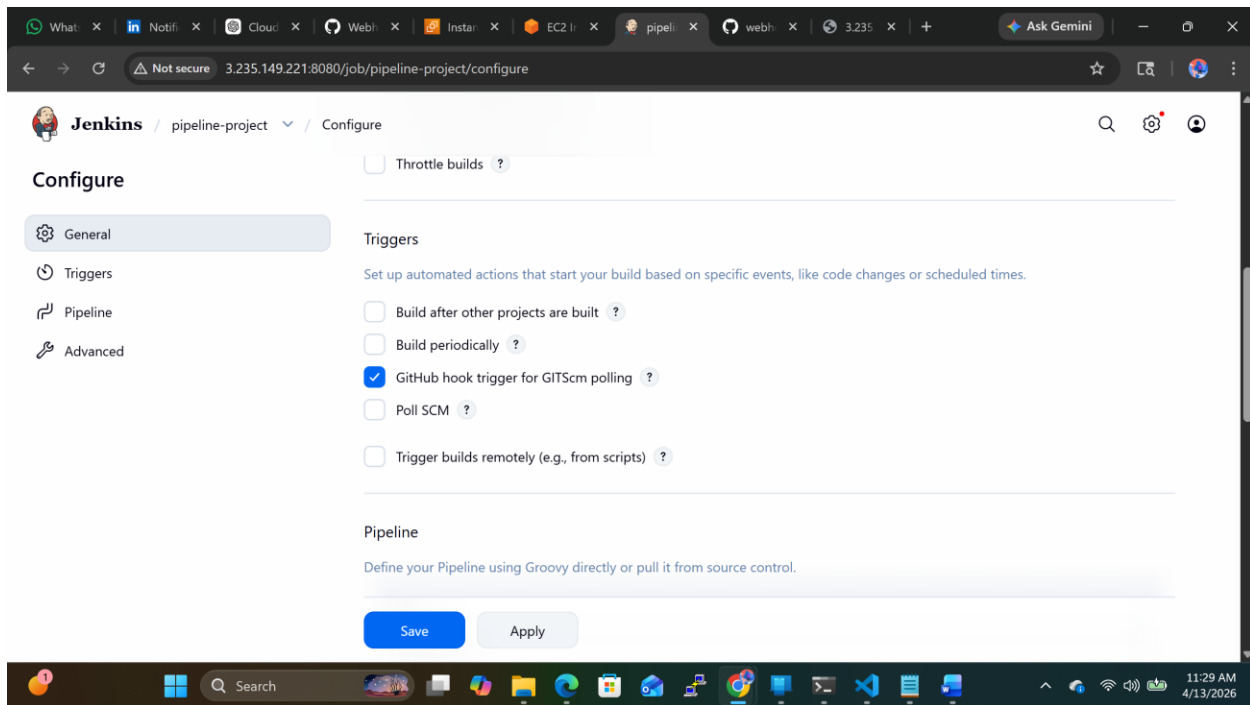
Project folder:



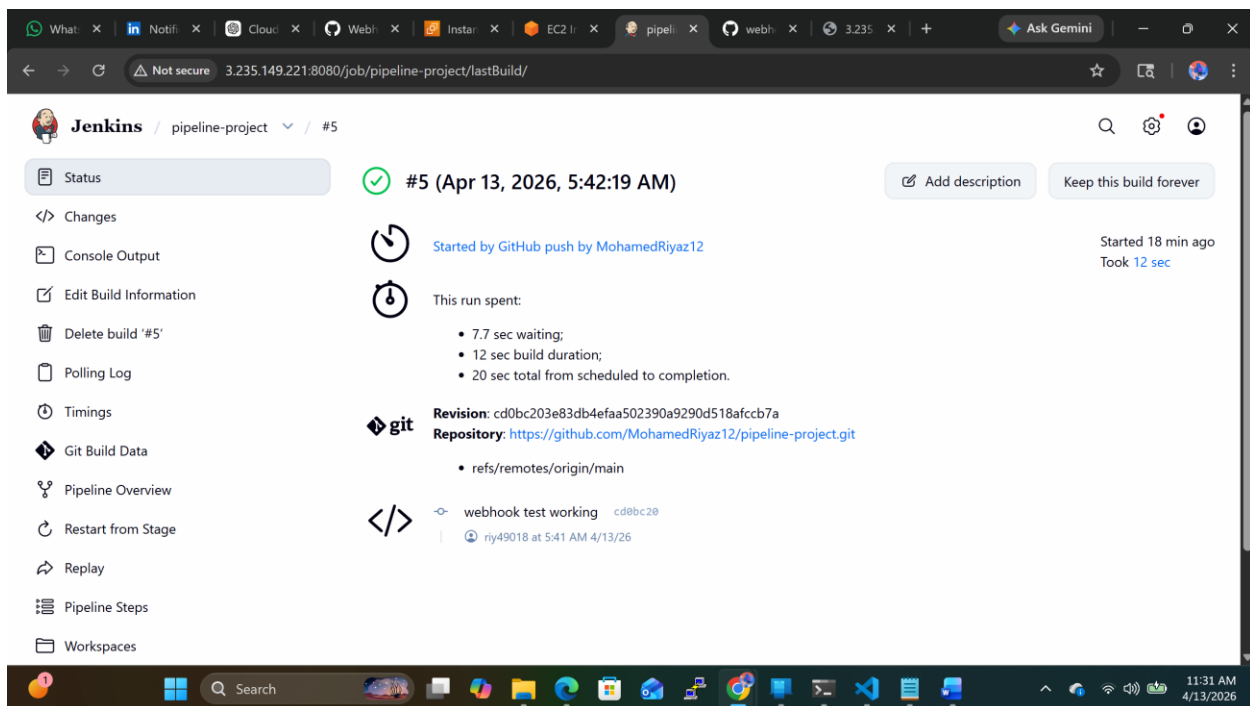
Files pushed to github repository:



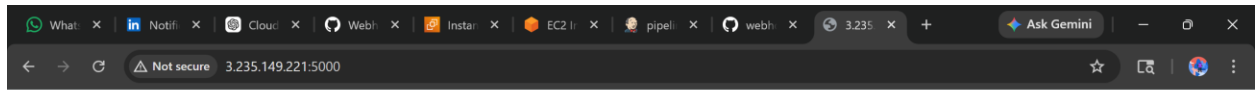
EC2 instance connected through AWS CLI:



Triggered run due to webhook:



Live app:



DevOps Project Running 🚀

